Referee-Style Report: *Matrix Theory Reloaded: A BPS Road to Holography*

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1. Principal achievement

The paper gives a unified target-space picture of the BPS decoupling limits that lead to matrix theories on Dp-branes (MpT), and uses it to build holography. Three points stand out:

- **Geometry.** The BPS limit (fine-tuning the RR potential so that brane tension equals charge) produces a flat or curved non-Lorentzian target space with a codimension-$(p+1)$ foliation and a p-brane boost symmetry. The paper derives this from a Lorentzian metric by rescaling with $\omega \to \infty$, then generalises it to curved backgrounds using vielbeins.
- **Holography.** Applying one such limit to the asymptotic region of a curved Dp-brane geometry reproduces the near-horizon limit in the bulk. Applying two limits in succession yields new holographic examples, some with non-Lorentzian bulk. This leads to the conjecture that holography is a DLCQ^n/DLCQ^m correspondence with $m > n$, with AdS$_5$/CFT$_4$ as the case DLCQ^0/DLCQ^1.
- $T\bar{T}$ **link.** Undoing the BPS limit appears as a p-brane generalisation of the $T\bar{T}$ deformation. In the Polyakov formulation this becomes a marginal $\lambda\bar{\lambda}$ current–current deformation. The paper derives flow equations for the Dp-brane DBI action, with the 3d flow for the D2/M2 case given as a clean example. This is a new and concrete way to view the emergence of the bulk geometry without a limiting procedure.

The synthesis is valuable. It connects matrix theory, non-relativistic string theory, non-Lorentzian holography and $T\bar{T}$ within one framework.

2. Principal drawback

The central holographic claims are not tested against equations of motion. The new non-Lorentzian bulk backgrounds (Section 5) are obtained by hand-rescaling the harmonic-function length scale ℓ with ω, which the paper interprets as a large-N rescaling. The authors state explicitly that they do not verify these geometries solve the equations of motion and torsional constraints of MpT supergravity, and that their supersymmetry and charges are not established. The DLCQ^n/DLCQ^m correspondence is a conjecture supported by examples, not a derivation. Several of the flow equations are obtained for abelian, bosonic D-brane actions, and the paper restricts its matrix-theory discussion to $p \leq 3$ because of strong-coupling issues. The result is a strong framework whose key new predictions still lack independent checks.

3. Most worthwhile follow-up (feasibility versus impact)

Proposal: check whether the rescaled near-horizon MpT backgrounds solve the MpT limit of type II supergravity, at least to leading order in $1/\omega$ and in a truncated bosonic sector (metric, dilaton, RR $(p+2)$-form, torsion constraints).

- **Why this one.** It targets the paper's most explicit open item. A positive result would make the non-Lorentzian bulk geometries credible, and a negative result would show which rescalings are inconsistent. Either outcome is informative for the DLCQ conjecture.
- **Feasibility.** High. The method is a symbolic computation. It requires the MpT supergravity equations, which the cited non-relativistic supergravity papers provide, and a computer algebra system (for example Cadabra, or SymPy or Mathematica for smaller cases). The starting point is the $p = 2$ or $p = 3$ example, where the computation stays tractable.
- **Impact.** Moderately high. It bears directly on the holography claims and on which non-Lorentzian holographic duals are physically allowed. It also tests the link between the rescaling and large-N physics that the paper leaves open.

A more ambitious alternative is to derive the non-abelian extension of the p-brane $T\bar{T}$ flow. This would have higher impact but lower feasibility, since no clear route to the non-abelian DBI action is known. I would pursue the supergravity consistency check first, since it can give a definite answer within a few months.